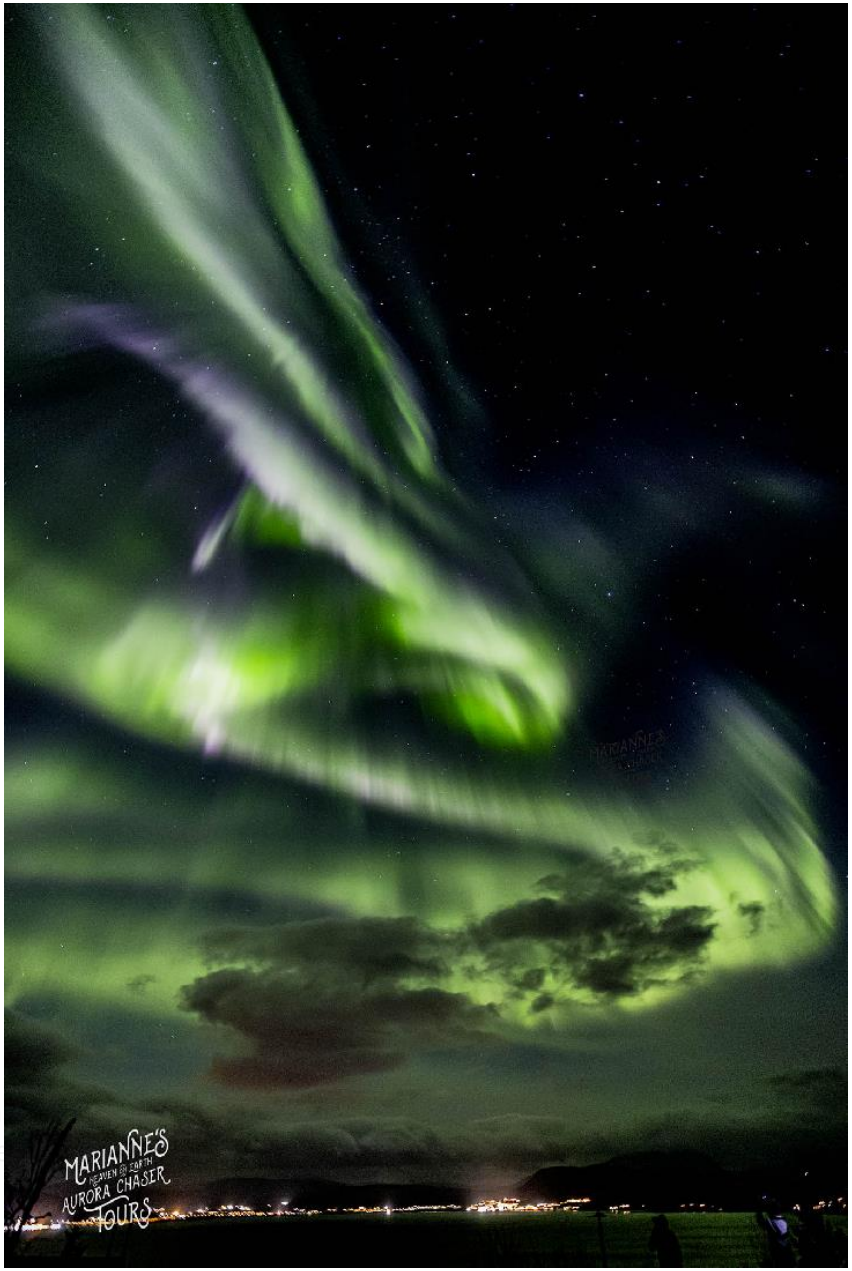
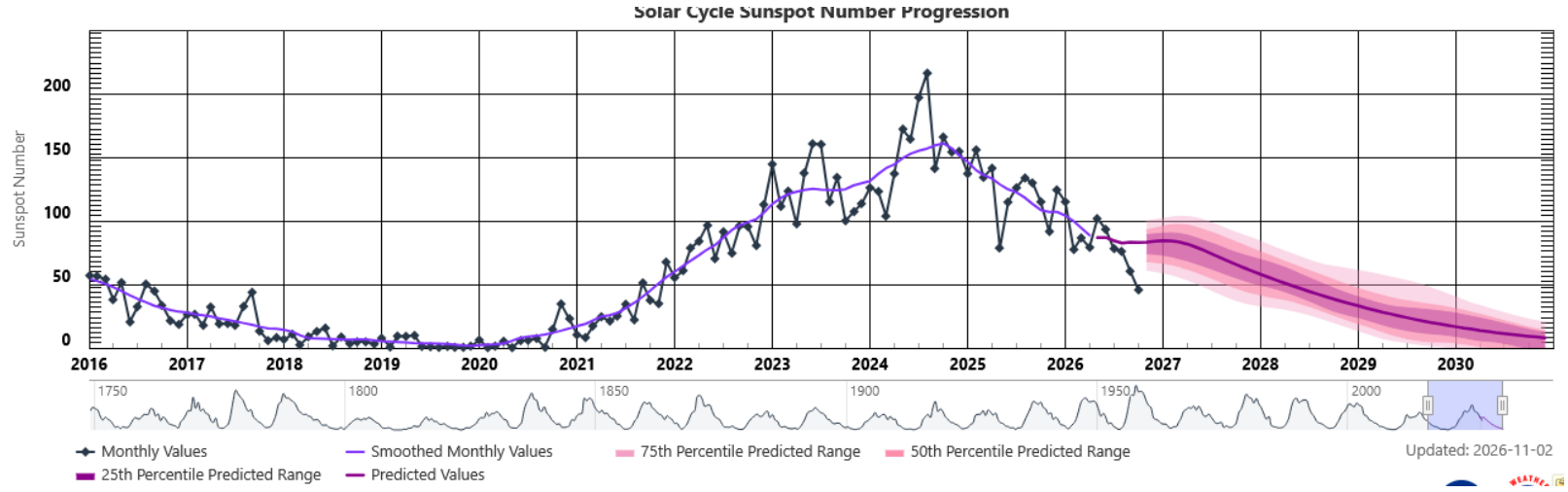
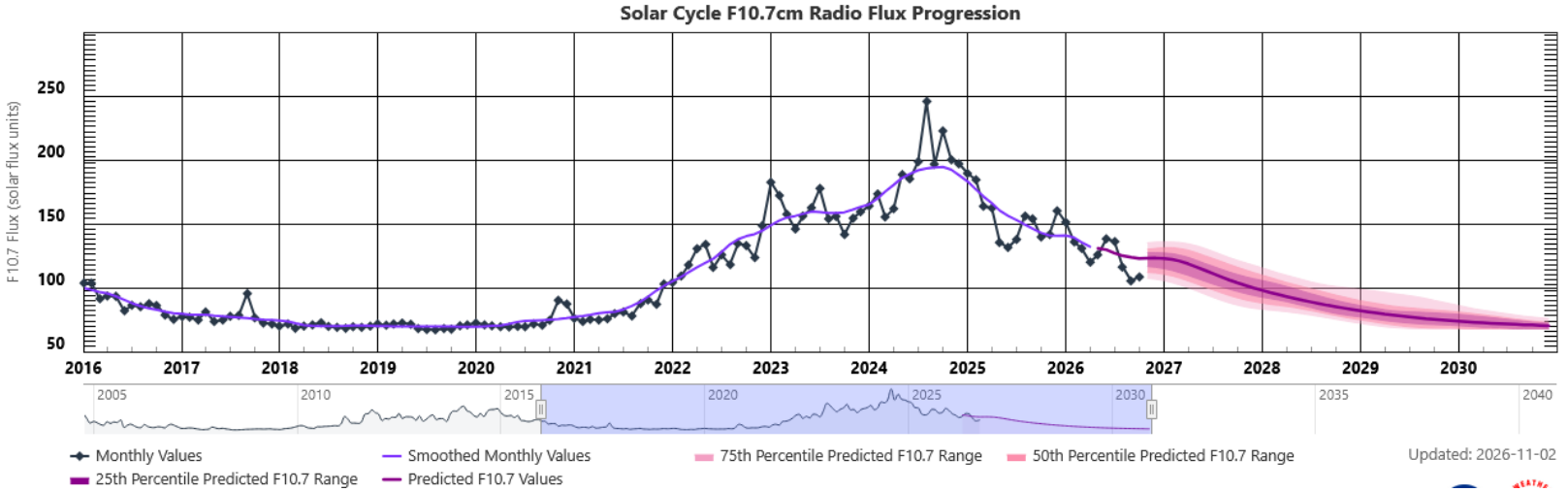


SOLAR WEATHER
3 NOV 2026

Lewis Thompson
W5IFQ

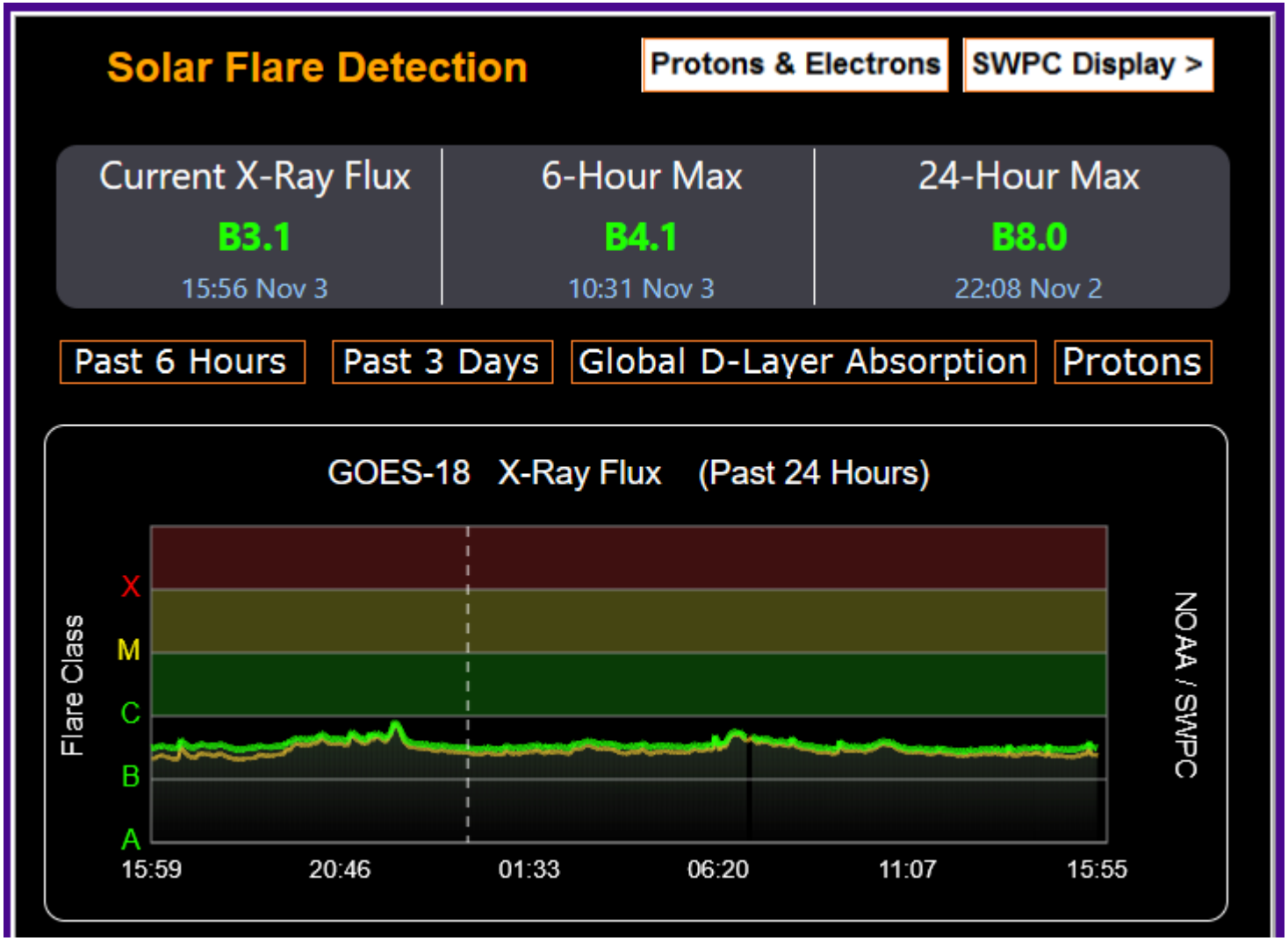
Taken by Marianne Bergli on October 31, 2026 @ Nordkjosbotn, Tromsø, Norway


Solar Cycle Sunspot Number Progression
Sunspot Number
200
150
100
50
0
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
2027
2028
2029
2030
1750
1800
1850
1900
1950
2000
Monthly Values
Smoothed Monthly Values
75th Percentile Predicted Range
50th Percentile Predicted Range
25th Percentile Predicted Range
Predicted Values
Updated: 2026-11-02


Solar Cycle F10.7cm Radio Flux Progression
F10.7 Flux (solar flux units)
250
200
150
100
50
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
2027
2028
2029
2030
2005
2010
2015
2020
2025
2030
2035
2040
Monthly Values
Smoothed Monthly Values
75th Percentile Predicted F10.7 Range
50th Percentile Predicted F10.7 Range
25th Percentile Predicted F10.7 Range
Predicted F10.7 Values
Updated: 2026-11-02

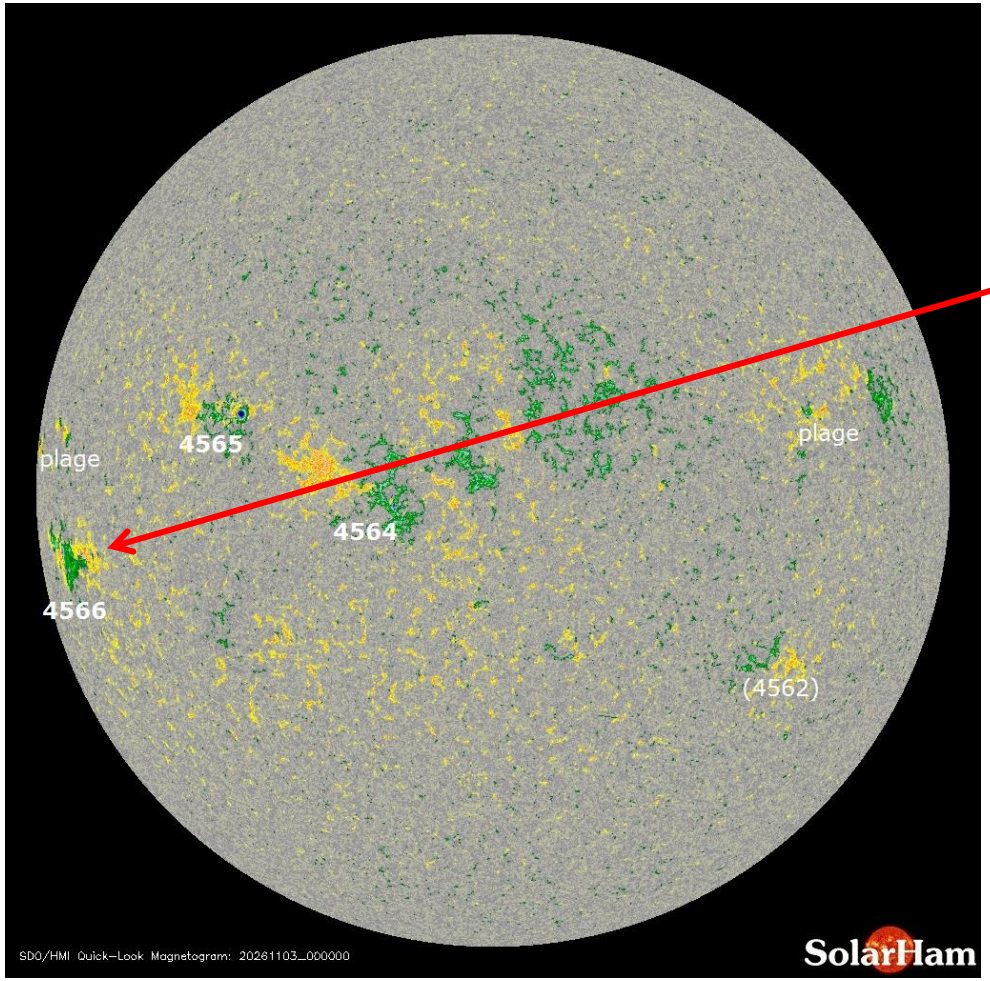

SolarHam
Indices: (11/03 @ 00:35 UTC)
SFI 97 ▲2 SSN 37
Geomagnetic Forecast
Nov. 3 Nov. 4 Nov. 5
2 (G0) 2 (G0) 3-4 (G0)
Max Kp
M-Lat 01% M-Lat 01% M-Lat 15%
H-Lat 20% H-Lat 20% H-Lat 40%
Probabilities
Detailed Forecast
Visible Sunspot Regions
AR 4566 B S07E66 —
AR 4565 A N13E35 —
AR 4564 A N03E11 —
Updated @ 00:35 UTC (November 3)
Solar Flare Probabilities
C-Flare: 30% M-Flare: 05%
X-Flare: 01% Proton: 01%
Probability Details
Updated @ 19:40 UTC (November 2)

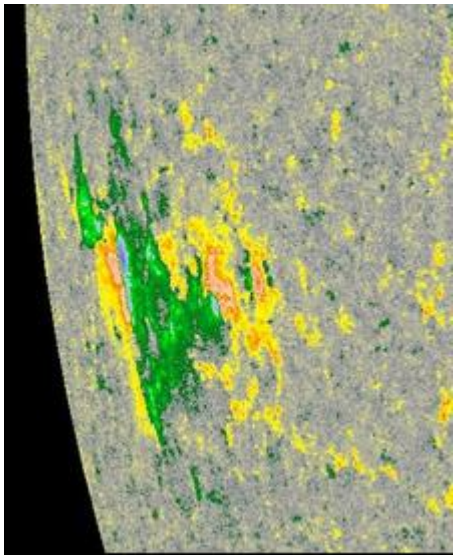
SolarHam


Solar Flare Detection
Protons & Electrons
SWPC Display >
Current X-Ray Flux
B3.1
15:56 Nov 3
6-Hour Max
B4.1
10:31 Nov 3
24-Hour Max
B8.0
22:08 Nov 2
Past 6 Hours
Past 3 Days
Global D-Layer Absorption
Protons
GOES-18 X-Ray Flux (Past 24 Hours)
Flare Class
X
M
C
B
A
15:59
20:46
01:33
06:20
11:07
15:55
NOAA / SWPC

Sun Spots

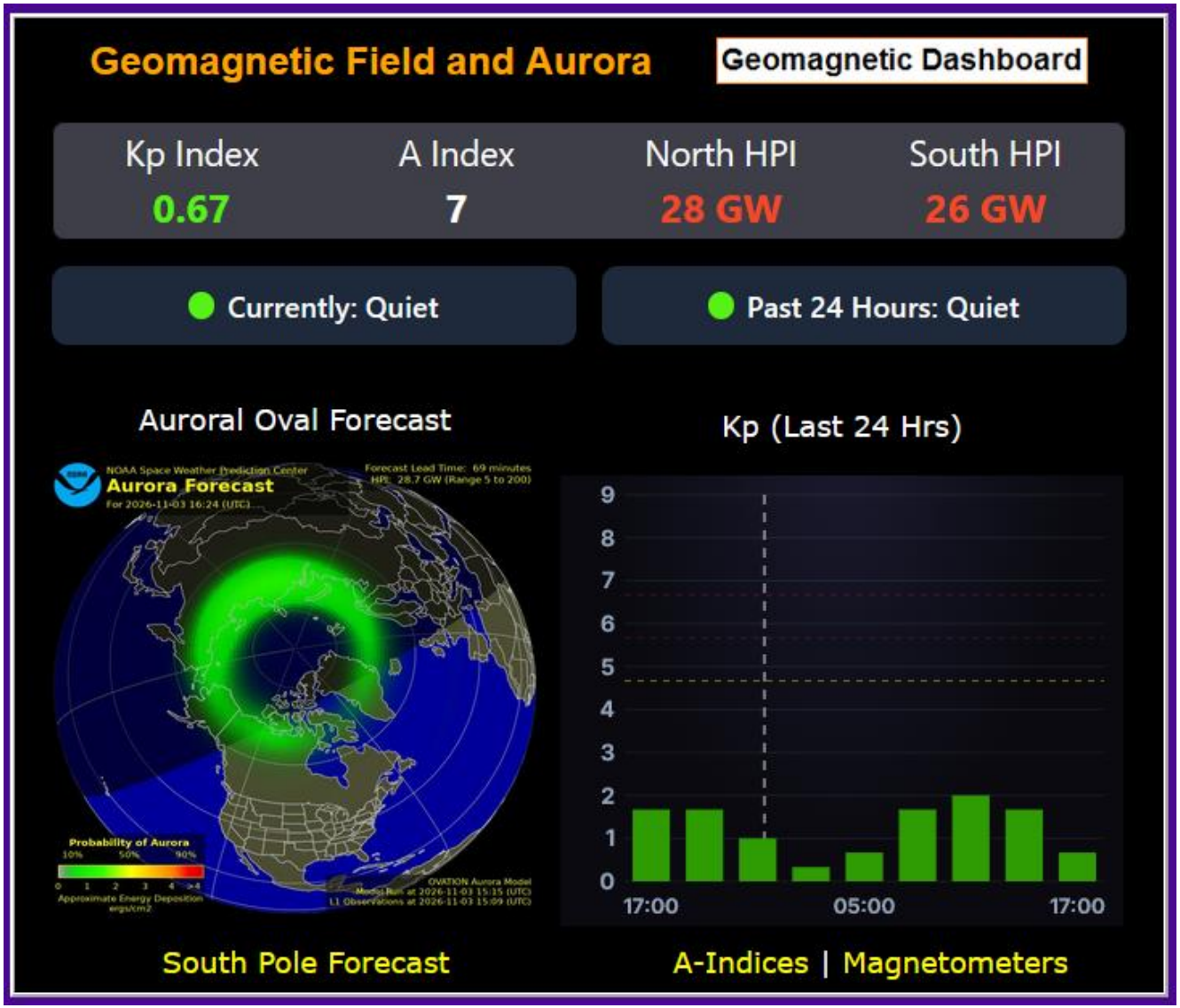
Magnetogram Image (Updated November 3, 2026)

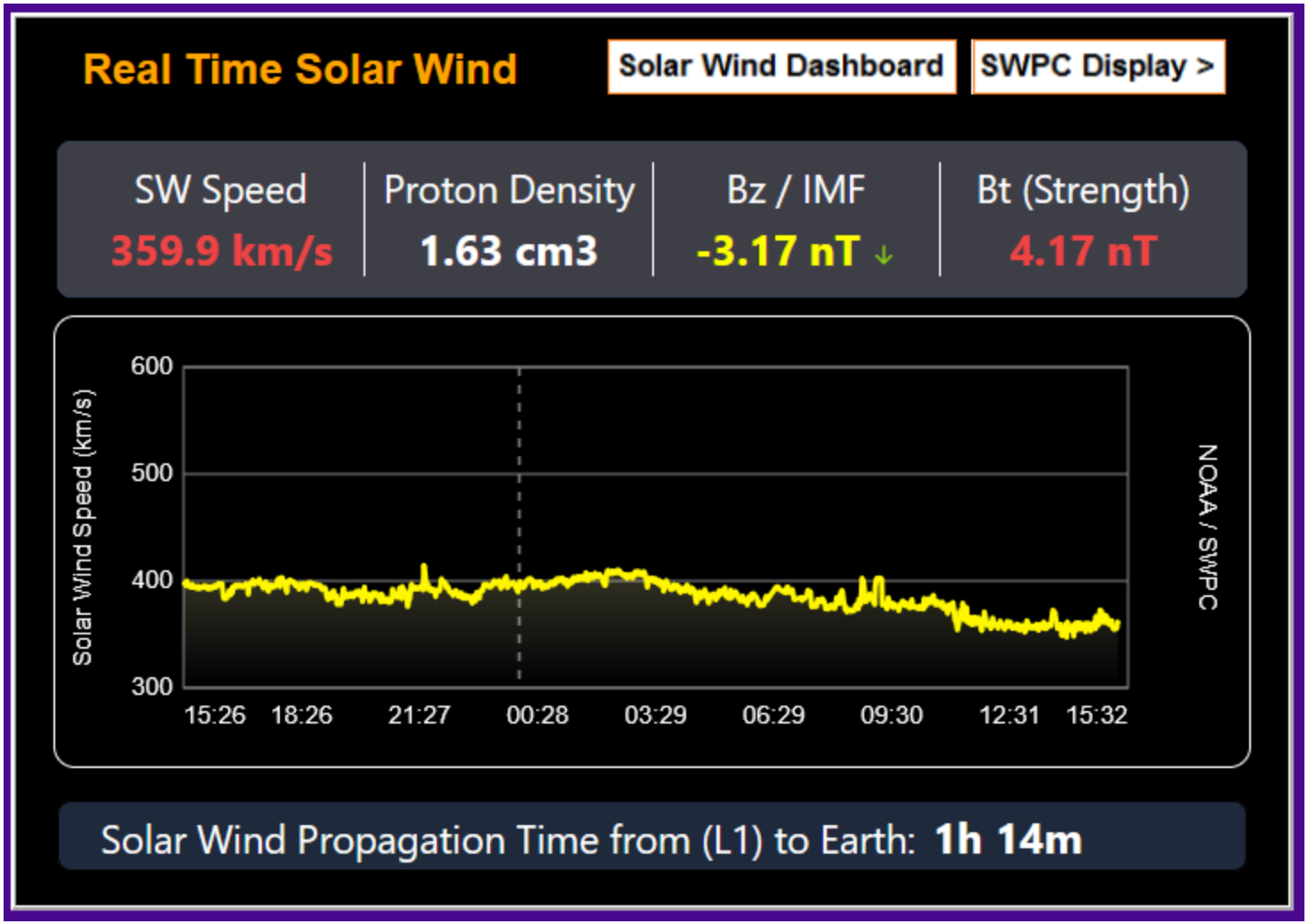




4566
Beta

Earth’s Geomagnetic Activity

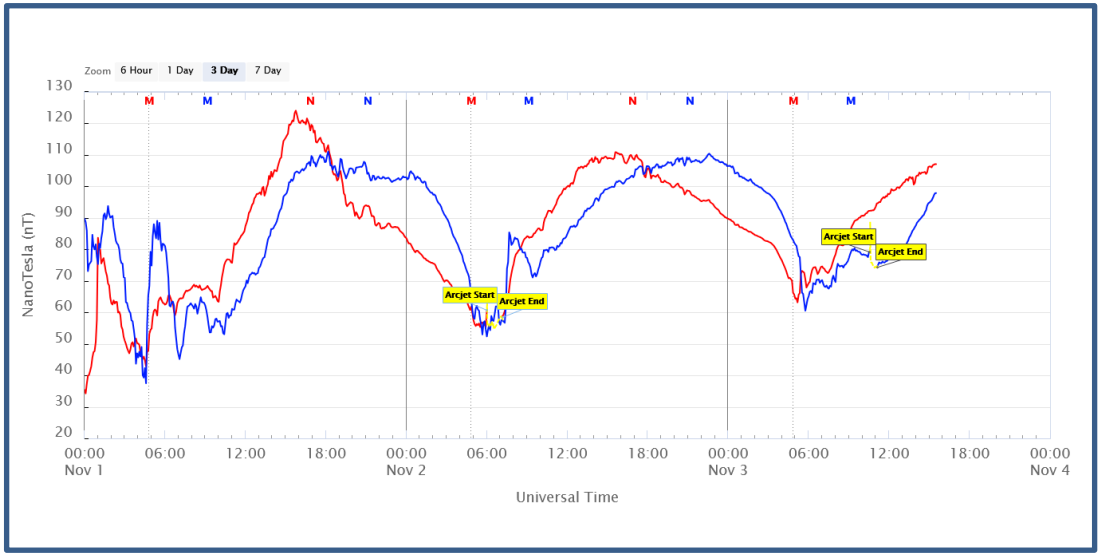



Real Time Solar Wind
Solar Wind Dashboard
SWPC Display >
SW Speed
359.9 km/s
Proton Density
1.63 cm3
Bz / IMF
-3.17 nT
Bt (Strength)
4.17 nT
Solar Wind Speed (km/s)
600
500
400
300
15:26
18:26
21:27
00:28
03:29
06:29
09:30
12:31
15:32
NOAA / SWPC
Solar Wind Propagation Time from (L1) to Earth: 1h 14m

Geomagnetic Conditions: 6 OCT 2026

Solar wind:
Bz = -3.17 nT
speed = 359.9 km/sec
density = 1.63 protons/cm^3

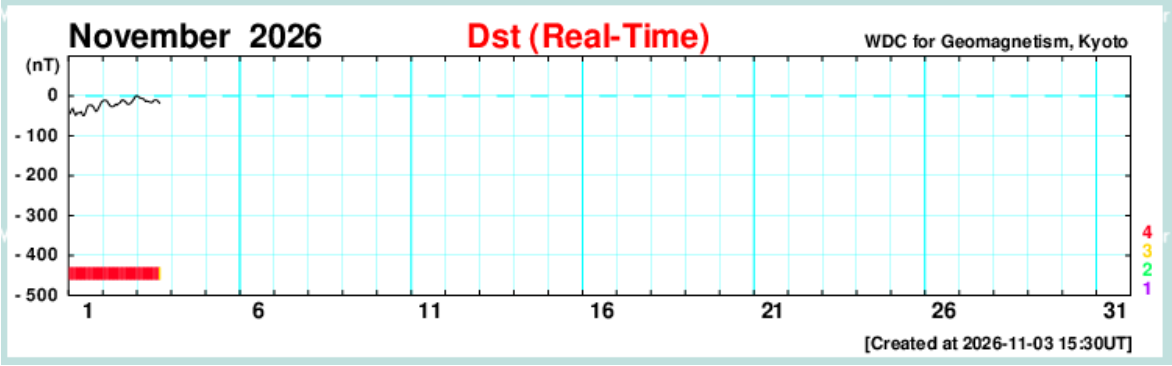
(From – NOAA DSCOVR
In L1, Lagrange Point)



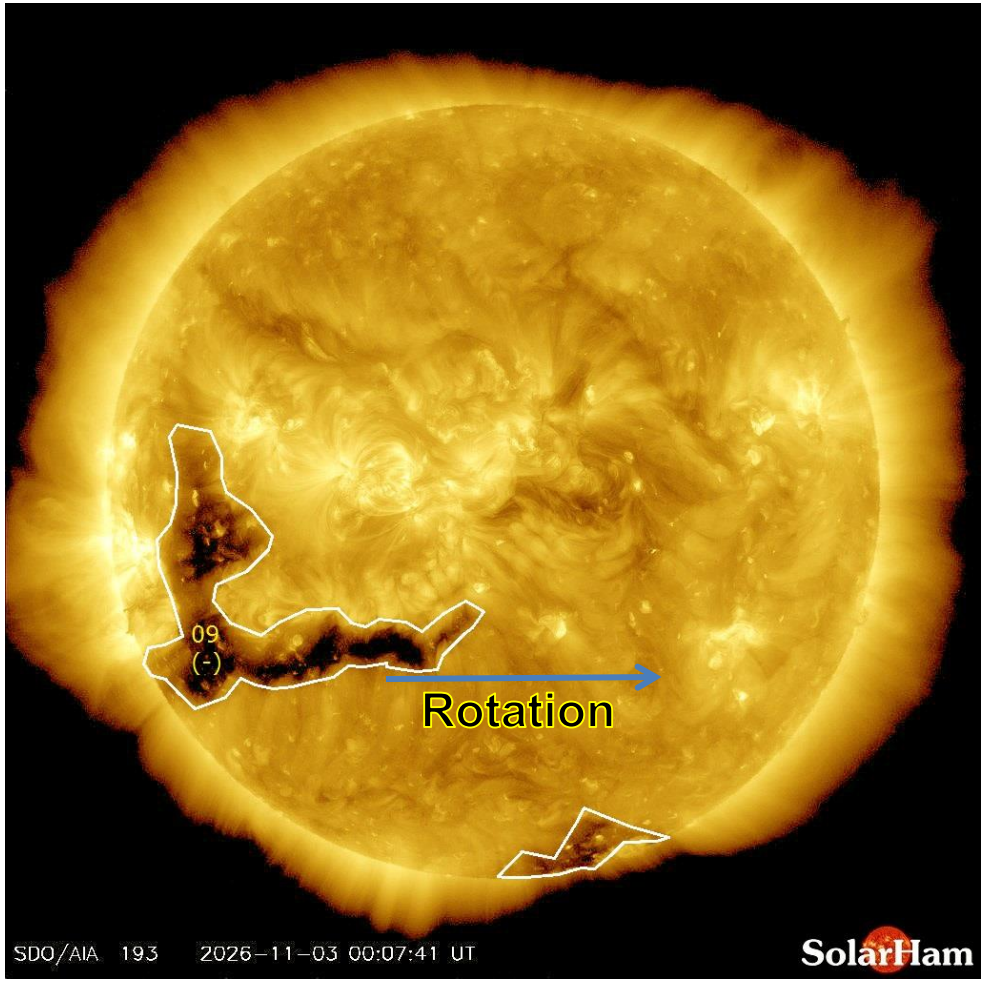
Dst = -19 nT (Ring Field)

(From – Data Analysis Center
For Geomagnetics and Space
Magnetism – Kyoto University)

From – GOES 18, 19
In geostationary orbit



Coronal Holes – 3- NOV 2026



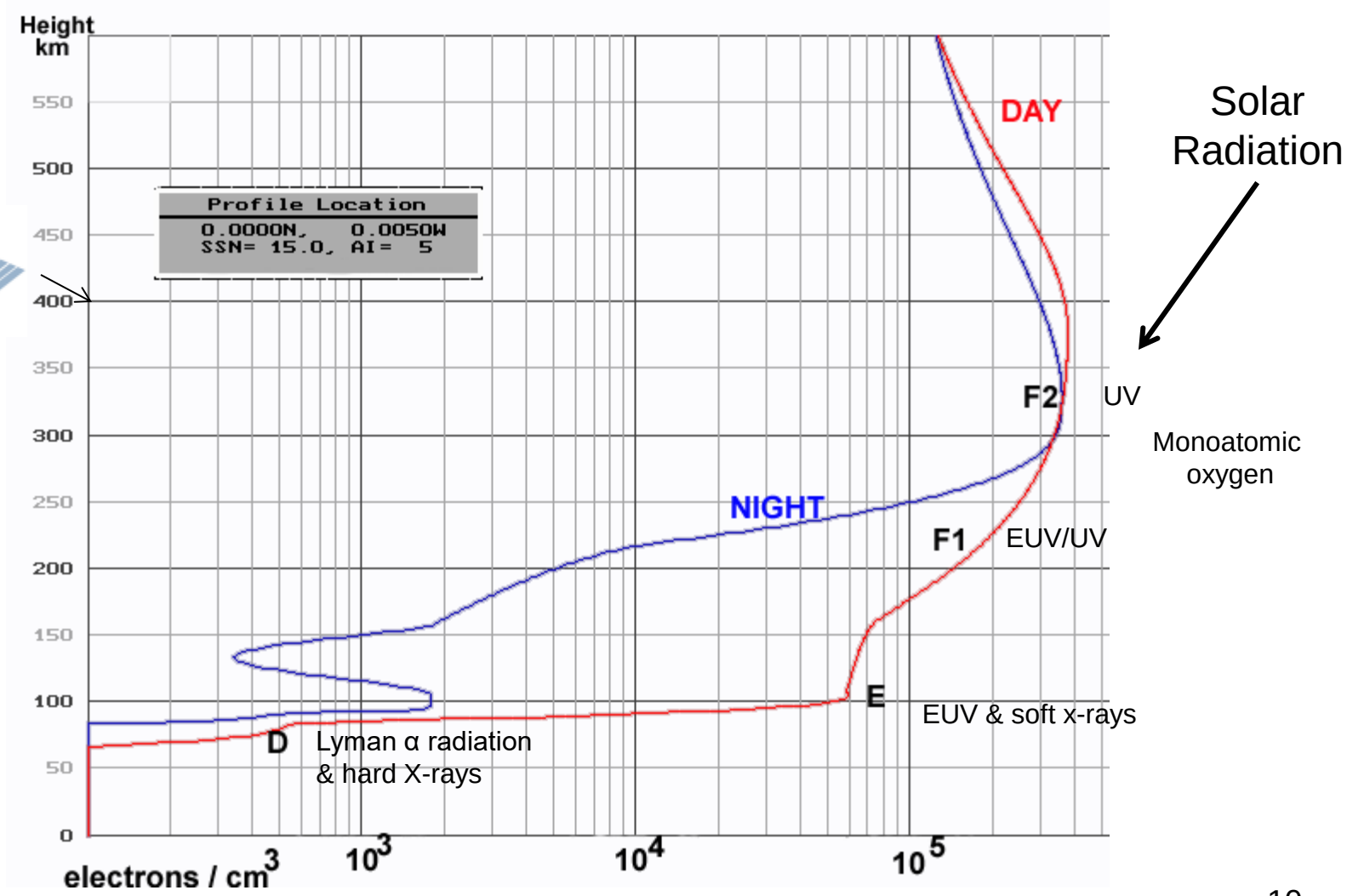
Analysis

Coronal hole #09 will be monitored as it rotates across the visible solar disk.

Ionosphere Creation

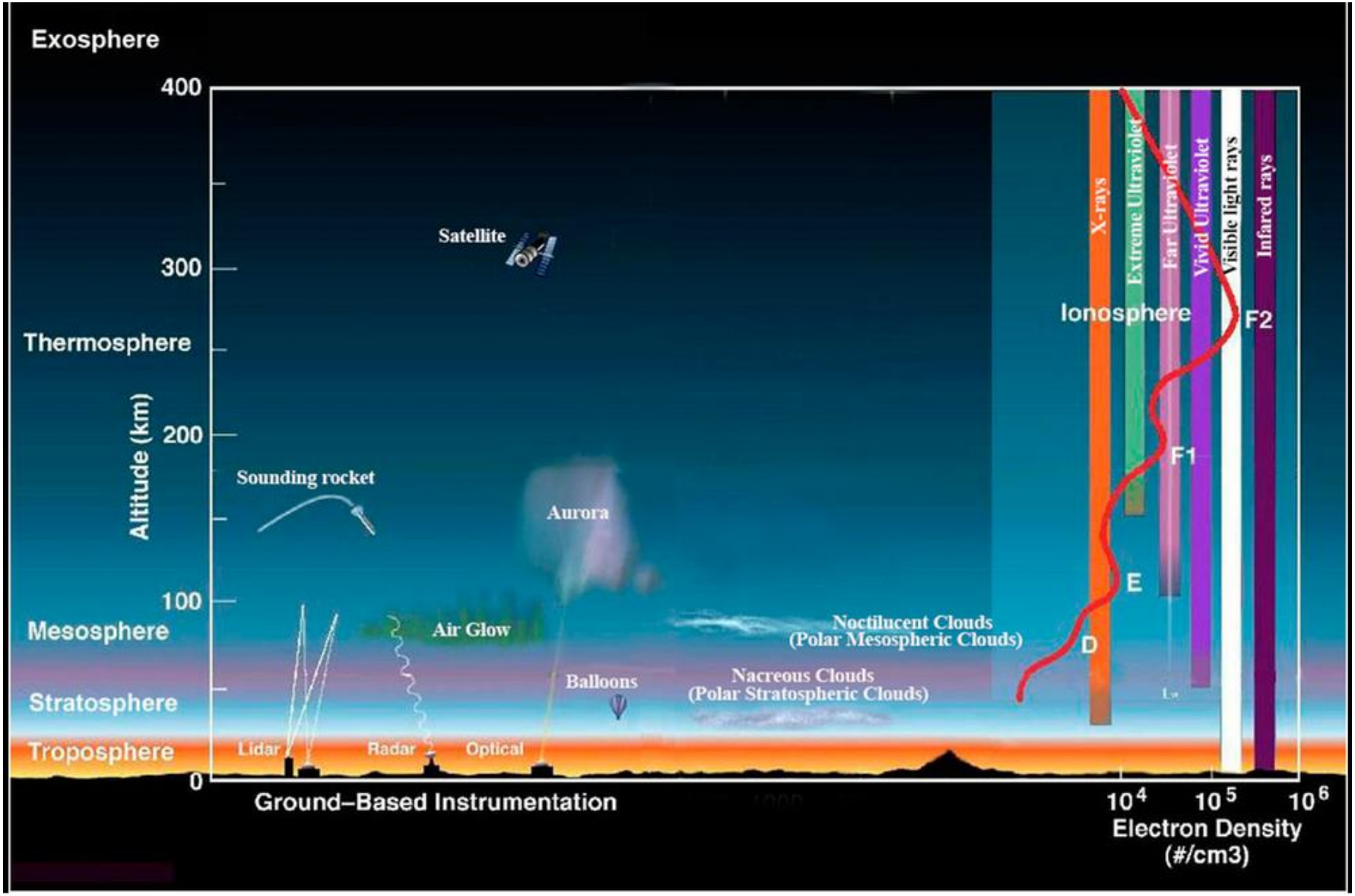


Gravity



Solar Radiation

Monoatomic oxygen

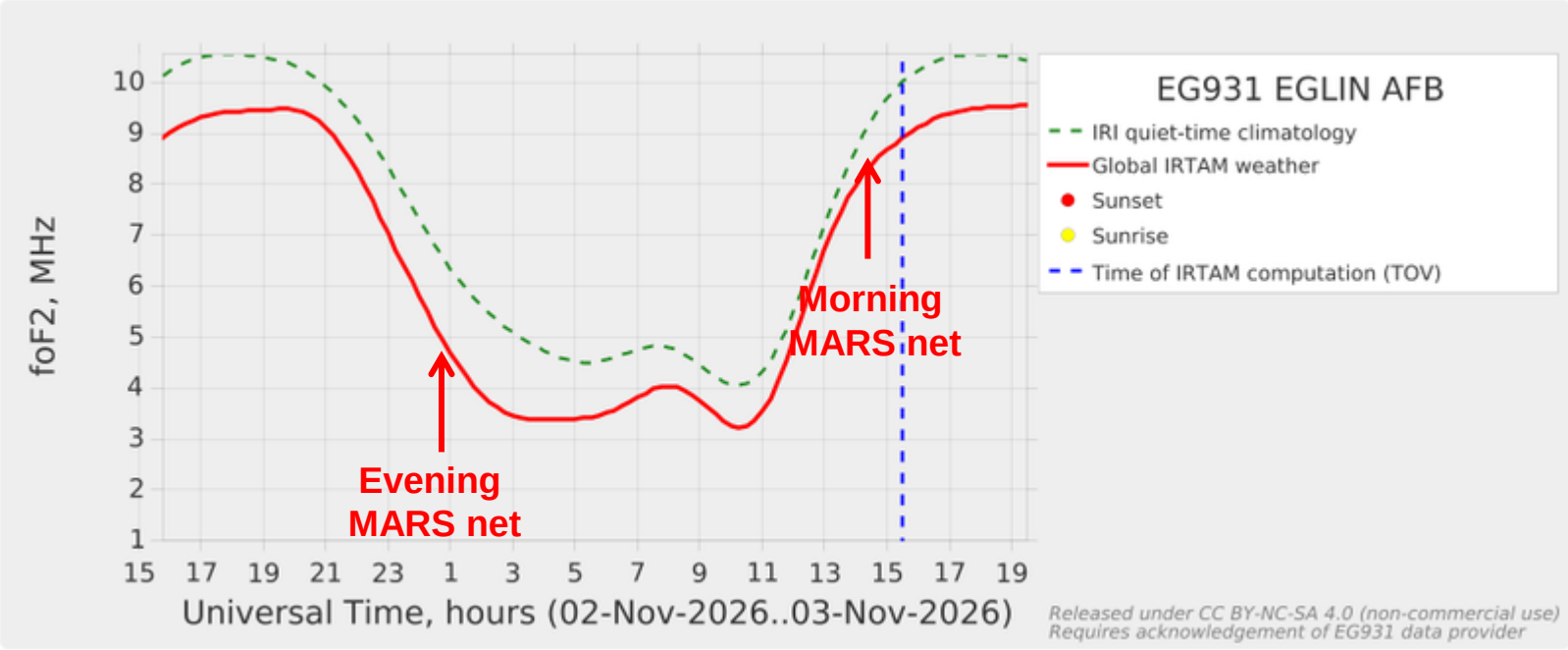

Exosphere
400
300
Thermosphere
200
100
Mesosphere
Stratosphere
Troposphere
0
Altitude (km)
Satellite
Sounding rocket
Aurora
Air Glow
Noctilucent Clouds
(Polar Mesospheric Clouds)
Balloons
Nacreous Clouds
(Polar Stratospheric Clouds)
Lidar
Radar
Optical
Ground–Based Instrumentation
X-rays
Extreme Ultraviolet
Far Ultraviolet
Vivid Ultraviolet
Visible light rays
Infared rays
Ionosphere
F2
F1
E
D
10^4
10^5
10^6
Electron Density
(#/cm3)

Austin Ionosonde Status

- GIRO is presently providing only GAMBIT models for both Austin and Eglin Ionosonde but these foF2 models appear to be accurate, based on MARS net performance (Critical Frequency).

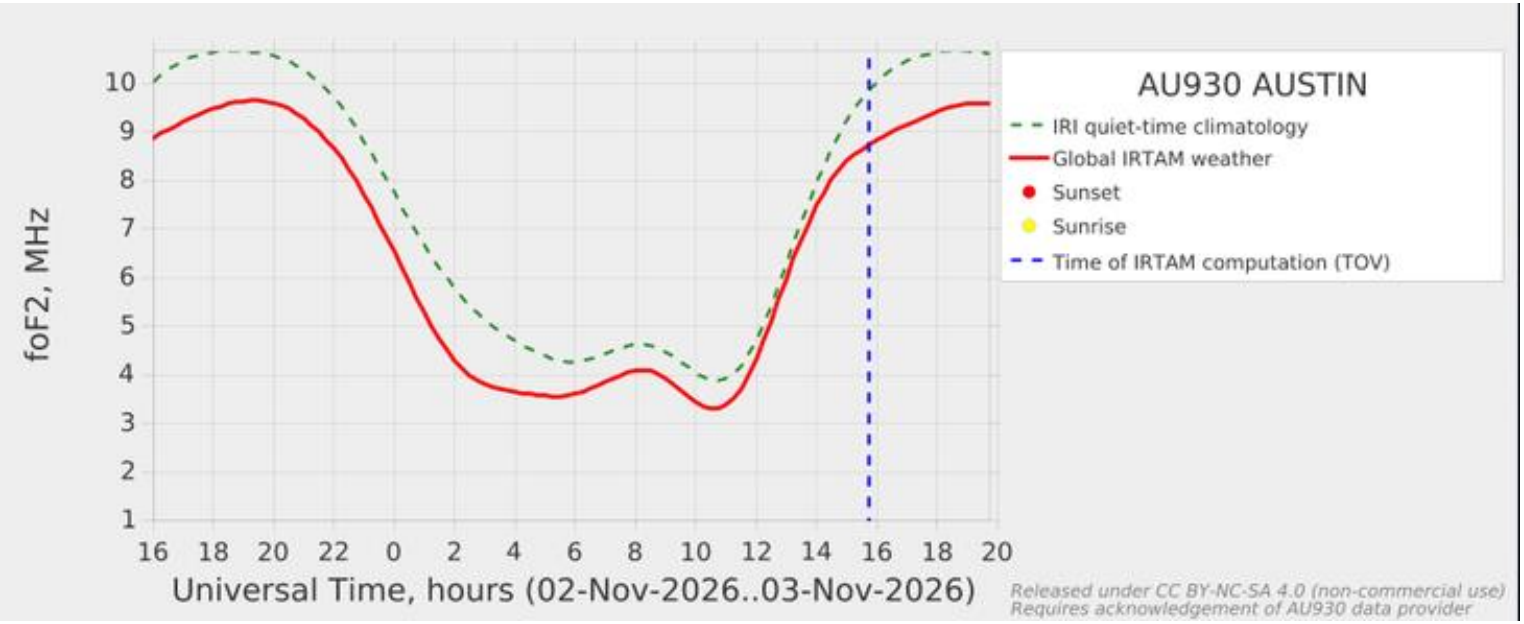
GAMBIT foF2 Trending Chart for Eglin Ionosonde

https://www.region6armymars.org/resources/solarweather.php



GAMBIT foF2 Trending Chart for Austin Ionosonde – Model only

https://www.region6armymars.org/resources/solarweather.php



Use of GIRO DIDBASE

https://giro.uml.edu/didbase/



Eglin AFB

Presently, Ionograms from Eglin are delayed by several days.

Time Shift for Eglin Ionosonde

- Eglin AFB Ionosonde is east of Austin at about the same Latitude so sun angle is about the same (same solar insolation).
- What happens to the Ionosphere over Eglin will occur over Austin about 45 minutes later.
- This is simply a Longitude difference converted to time.
- So look at Eglin's foF2 45 minutes earlier to see what is going to happen over Austin.
- The most rapid change in foF2 occurs as the sun is rising and setting (morning and evening).

NOTABLE SOLAR WX EVENTS HISTORIC

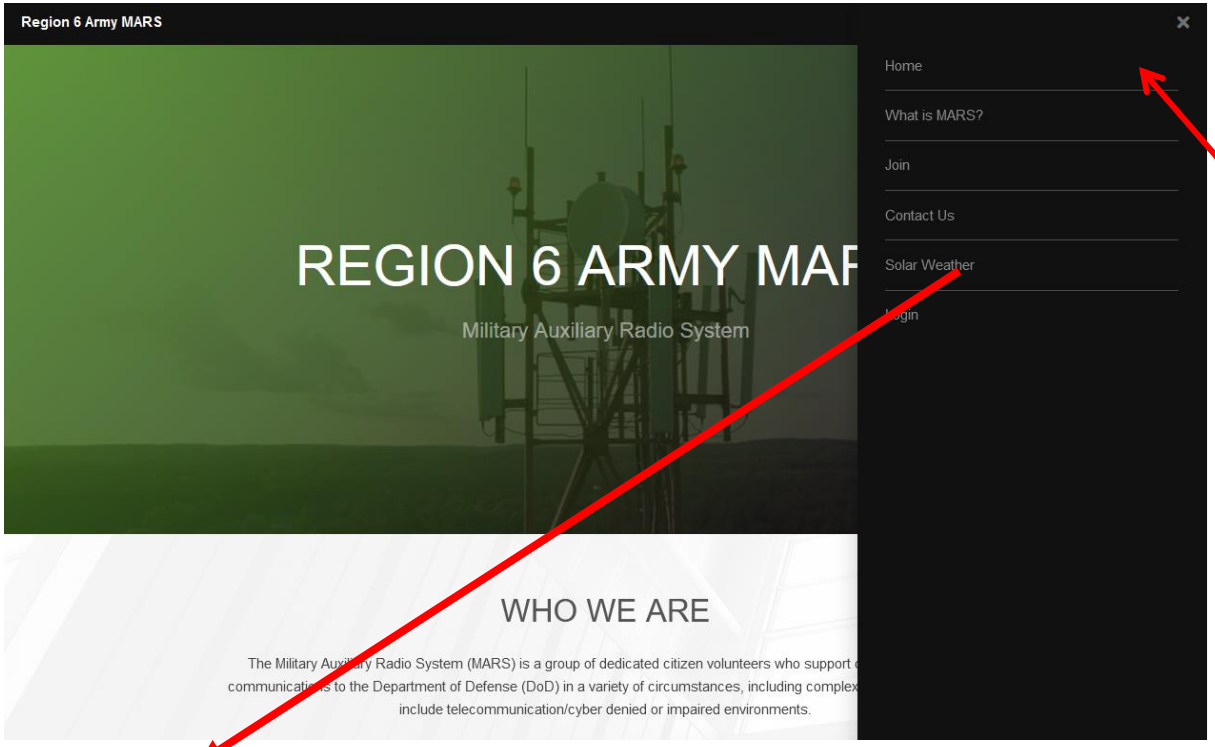
23 Years Ago / Great Halloween Storms

October 29, 2026 @ 15:00 UTC

While a minor (G1) geomagnetic storm is forecast for this weekend on Halloween, it will be nothing compared to what occurred 23 years ago this week during the height of Solar Cycle 23. A massive sunspot cluster numbered 486 unleashed several major solar flares, including this X17 explosion on October 28, 2003. This hurled an energetic and fast moving CME that struck Earth`s geomagnetic field less than 24 hours later. This was part of a series of events known as the great Halloween solar storms which included a long lasting extreme (G5) geomagnetic storm and worldwide displays of aurora. On November 4, 2003 while no longer in view off the west limb, 486 produced a historic solar flare that was estimated to measure X28+. Great memories from over two decades ago!



Solar Weather Data



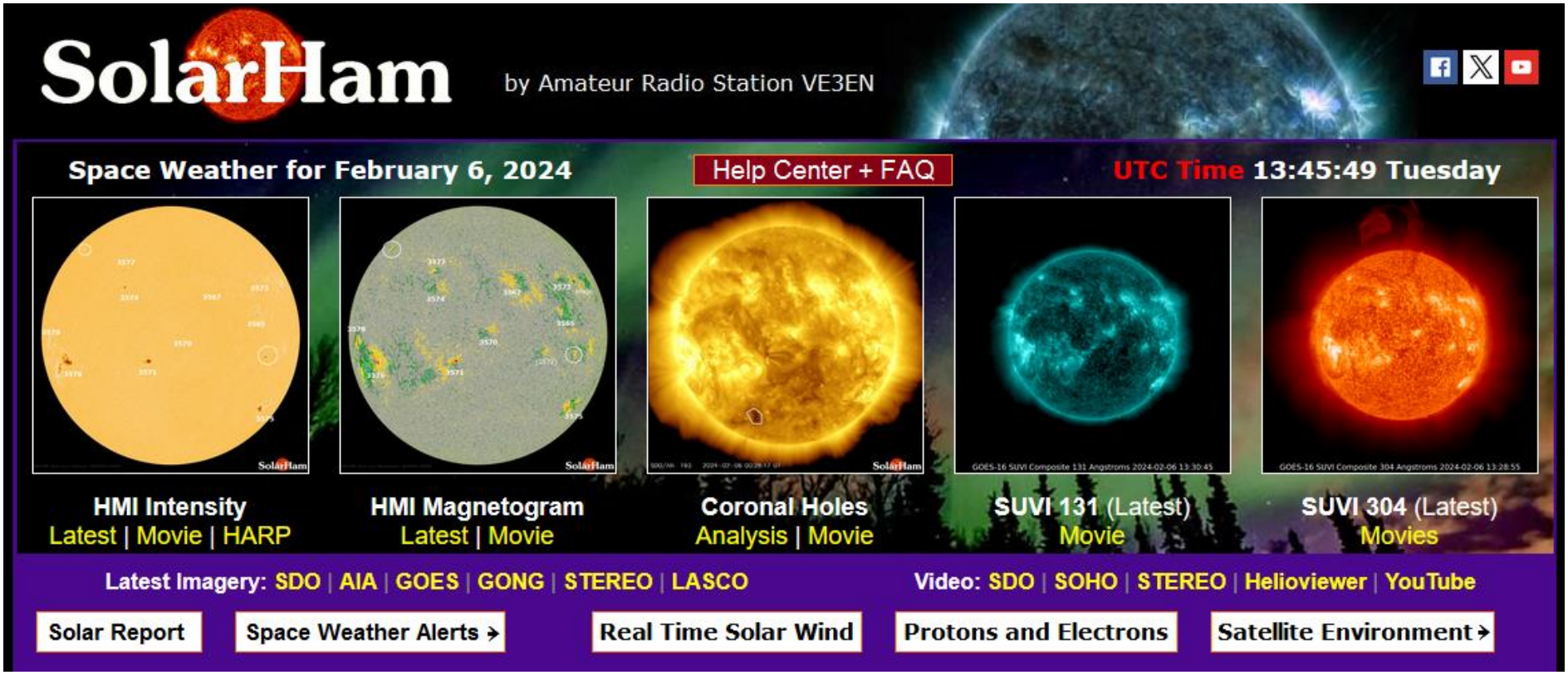
Menu

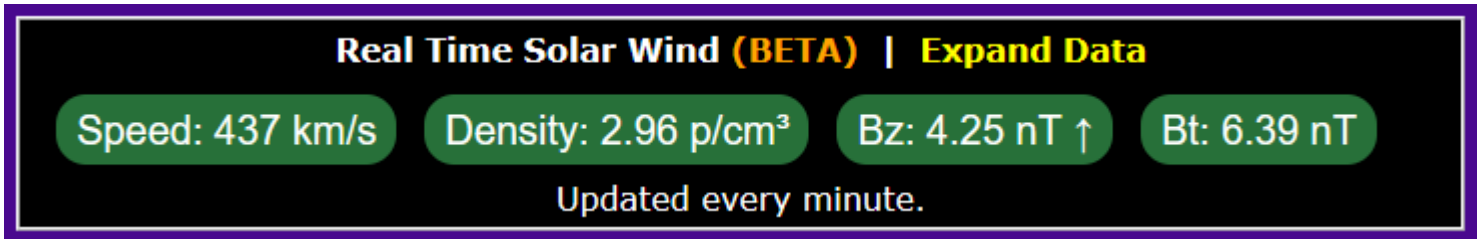
Solar Weather

All Ionosondes_
GAMBIT URL

- GAMBIT - Global Assimilative Model of Bottomside Ionosphere Timeline
  - Boulder
  - Eglin

- NOAA Solar Weather - Solar Weather plots of Kp and X-Ray and other solar emissions.
- Solen Solar Weather - Good general solar forecast from an individual.
- Solar Ham - SolarHam provides real time solar news, as well as consolidated data from various sources.


SolarHam
by Amateur Radio Station VE3EN
Space Weather for February 6, 2024
Help Center + FAQ
UTC Time 13:45:49 Tuesday
HMI Intensity
Latest | Movie | HARP
HMI Magnetogram
Latest | Movie
Coronal Holes
Analysis | Movie
SUVI 131 (Latest)
Movie
SUVI 304 (Latest)
Movies
Latest Imagery: SDO | AIA | GOES | GONG | STEREO | LASCO
Video: SDO | SOHO | STEREO | Helioviewer | YouTube
Solar Report
Space Weather Alerts
Real Time Solar Wind
Protons and Electrons
Satellite Environment


Real Time Solar Wind (BETA) | Expand Data
Speed: 437 km/s
Density: 2.96 p/cm³
Bz: 4.25 nT ↑
Bt: 6.39 nT
Updated every minute.

spaceweather.com

News and information about the Sun-Earth environment

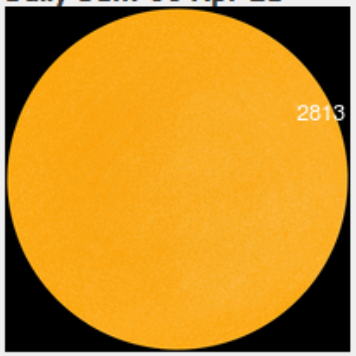
https://www.spaceweather.com/

Current Conditions

Solar wind
speed: **314.8** km/sec
density: **9.9** protons/cm^3
more data: ACE, DSCOVR
Updated: Today at 1225 UT

X-ray Solar Flares
6-hr max: **A1** 1027 UT Apr06
24-hr: **A1** 1515 UT Apr05
explanation | more data
Updated: Today at: 1230 UT

Daily Sun: 06 Apr 21



Sunspot AR2813 is decaying, and poses no threat for strong flares.
Credit: SDO/HMI

FLYING TO THE VOLCANO: Iceland's Geldingadalur volcano has turned into an popular tourist attraction---especially since auroras were sighted above the glowing lava. Early this morning, Tuesday, April 6th, Brian Emfinger saw auroras before he even reached the Reykjanes peninsula:



QUESTIONS?

Lewis Thompson

W5IFQ@att.net

512-587-9944